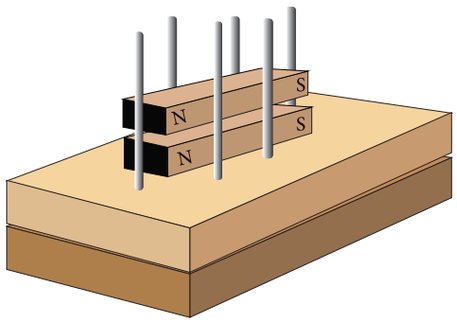
1.11 Science Today – Bullet Trains



We Know that Like poles of the magnet repel each other. Keep two Bar magnets as shown in the Figure.

What do you observe? _____

By using repulsion we can levitate a magnetic object. Let us make a toy and enjoy magnetic levitation.

Levitating propeller


Ring magnets

Bottle propeller




Plastic bottle filled with sand


Tape


Scissors


Stiff plastic straw

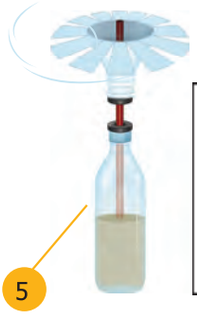
1 Make a propeller from a 500-ml plastic bottle. Make a hole in the bottle lid.



2 Screw the lid with the hole on a bottle half filled with sand. Press fit a stiff straw in the lid. Embed the straw in the sand to make it stand erect.







Place a few ring magnets in the straw. Similar poles will repel each other.

4 Like poles repel and this levitates the bottle fan. The ceiling fan makes it spin.





3 Place two magnets each inside and outside the propellor lid.These magnets will automatically stick to each other.

5 Place the propeller on the stiff straw.Ther magnets in the straw and the propeller should repel each other. This will make the propeller levitate. On placing it under a ceiling fan the propeller will spin very fast!



Have you enjoyed with this toy? Electromagnetic train is working in the same principle. Have you heard about it?

Electromagnetic train is called as suspension train and also called as flying train. It does not require diesel or petrol. This technology uses the property of magnetic attraction and repulsion to run these super fast electromagnetic trains.

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